1. **Schema Validation**
   Objective: Verify that all tables and constraints were created correctly.

Sql Commands used:

```
SHOW TABLES;
DESCRIBE users;
DESCRIBE students;
DESCRIBE instructors;
DESCRIBE courses;
DESCRIBE enrollments;
DESCRIBE audit_logs;
```

Screenshots:

Your SQL query has been executed successfully.

```
SHOW TABLES;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Tables_in_student_management_systemdb
audit_logs
courses
enrollments
instructors
students
users

Your SQL query has been executed successfully.

```
DESCRIBE users;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
userID	int(11)	NO	PRI	*NULL*	auto_increment
username	varchar(50)	NO	UNI	*NULL*	
password	varchar(255)	NO		*NULL*	
role	enum('admin','staff')	NO		*NULL*	

Your SQL query has been executed successfully.

```
DESCRIBE students;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
studentID	int(11)	NO	PRI	*NULL*	auto_increment
firstName	varchar(50)	YES		*NULL*	
lastName	varchar(50)	YES		*NULL*	
email	varchar(100)	YES	UNI	*NULL*	
created_at	timestamp	NO		current_timestamp()	

Your SQL query has been executed successfully.

```
DESCRIBE instructors;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
instructorID	int(11)	NO	PRI	*NULL*	auto_increment
name	varchar(100)	YES		*NULL*	
email	varchar(100)	YES	UNI	*NULL*	

Your SQL query has been executed successfully.

```
DESCRIBE courses;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
courseID	int(11)	NO	PRI	NULL	auto_increment
courseName	varchar(100)	YES		NULL	
instructorID	int(11)	YES	MUL	NULL	

Your SQL query has been executed successfully.

```
DESCRIBE enrollments;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
enrollment_id	int(11)	NO	PRI	NULL	auto_increment
studentID	int(11)	YES	MUL	NULL	
courseID	int(11)	YES	MUL	NULL	
enrollmentDate	date	YES		NULL	
grade	varchar(5)	YES		NULL	

Your SQL query has been executed successfully.

```
DESCRIBE audit_logs;
```

[Edit inline] [Edit] [Create PHP code]

Extra options

Field	Type	Null	Key	Default	Extra
logID	int(11)	NO	PRI	NULL	auto_increment
action	varchar(50)	YES		NULL	
tableName	varchar(50)	YES		NULL	
actionTime	timestamp	NO		current_timestamp()	

Result:
All tables were successfully created with correct columns, data types, primary keys, and foreign key constraints.

2. **Data Population (100,000+ Records)**
Objective: Ensure that a large dataset was successfully inserted.

Sql Commands Used:

```
SELECT
    (SELECT COUNT(*) FROM users) +
    (SELECT COUNT(*) FROM students) +
    (SELECT COUNT(*) FROM instructors) +
    (SELECT COUNT(*) FROM courses) +
    (SELECT COUNT(*) FROM enrollments) +
    (SELECT COUNT(*) FROM audit_logs)
AS total_records;
```

Screenshots:

Your SQL query has been executed successfully.

```
SELECT (SELECT COUNT(*) FROM users) + (SELECT COUNT(*) FROM students) + (SELECT COUNT(*) FROM instructors) + (SELECT COUNT(*) FROM
courses) + (SELECT COUNT(*) FROM enrollments) + (SELECT COUNT(*) FROM audit_logs) AS total_records;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

total_records
102000

Result:
The query result shows that the total number of records across all tables exceeds 100,000 rows. This confirms that the database has been successfully populated with a large dataset and meets the project requirement. The system is capable of handling high-volume data, simulating a real-world application environment.

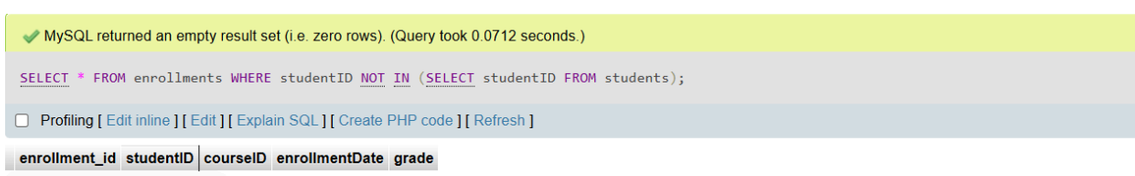
3. **Data Integrity Validation**
Objective: Ensure that all inserted data maintains referential integrity and contains no orphan records.

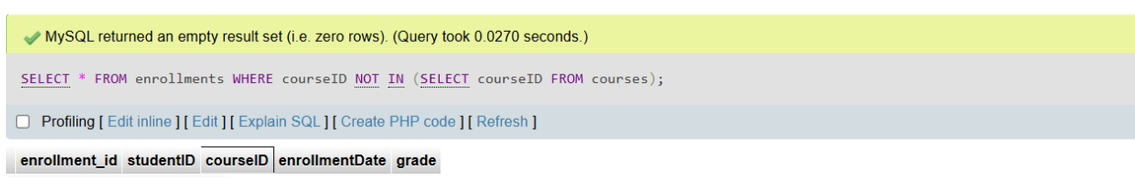
SQL Commands Used:

```
SELECT *
FROM enrollments
WHERE studentID NOT IN (SELECT studentID FROM students);

SELECT *
FROM enrollments
WHERE courseID NOT IN (SELECT courseID FROM courses);
```

Screenshots:





Result:
The queries returned zero rows, confirming that there are no orphan records. All foreign key relationships are valid, and the database maintains referential integrity.

4. **Relationship Testing (JOIN Validation)**
Objective: Verify that relationships between tables return correct and meaningful data.

SQL Commands Used:

```
SELECT
    s.firstName,
    s.lastName,
    c.courseName,
    e.grade
FROM enrollments e
JOIN students s ON e.studentID = s.studentID
JOIN courses c ON e.courseID = c.courseID
LIMIT 10;
```

Screenshots:

Showing rows 0 - 9 (10 total, Query took 0.0066 seconds.)

```
SELECT s.firstName, s.lastName, c.courseName, e.grade FROM enrollments e JOIN students s ON e.studentID = s.studentID JOIN courses c ON e.courseID = c.courseID LIMIT 10;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

firstName	lastName	courseName	grade
David	Rodriguez	Course 1	F
Yvette	Castillo	Course 1	C
Edward	Martin	Course 2	F
Mary	Juarez	Course 2	A
Bryan	Mendez	Course 2	B
Billy	Singh	Course 2	C
Brenda	Day	Course 2	C
Derek	Hogan	Course 2	B
Matthew	Calhoun	Course 3	A
Jeffrey	Davis	Course 3	B

Result:
The query successfully retrieved related data from multiple tables, confirming that relationships are properly implemented and functioning correctly.

5. **Query Performance Testing**
Objective: Ensure that the system can handle queries efficiently even with large datasets.

SQL Commands Used:

```
SELECT * FROM enrollments LIMIT 100;

SELECT COUNT(*) FROM enrollments;
```

Screenshots:

Showing rows 0 - 99 (100 total, Query took 0.0005 seconds.)

```
SELECT * FROM enrollments LIMIT 100;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

←T→	enrollment_id	studentID	courseID	enrollmentDate	grade
Edit Copy Delete	1	26268	4252	1985-12-07	D
Edit Copy Delete	2	31099	7259	1991-01-10	D
Edit Copy Delete	3	27213	1216	2026-01-17	D
Edit Copy Delete	4	24498	2573	1993-08-04	B
Edit Copy Delete	5	13605	1300	1974-05-16	B
Edit Copy Delete	6	24580	3793	2009-06-23	B
Edit Copy Delete	7	21509	9096	1998-07-08	A
Edit Copy Delete	8	7933	490	1990-12-26	F
Edit Copy Delete	9	35194	8352	1986-07-01	F
Edit Copy Delete	10	15475	6159	1979-09-18	A
Edit Copy Delete	11	35636	2621	1996-09-01	A
Edit Copy Delete	12	2097	2275	1979-01-17	A
Edit Copy Delete	13	24557	6342	2002-12-27	B

	enrollment_id	studentID	courseID	enrollmentDate	grade
Edit Copy Delete	14	24277	413	2009-09-17	C
Edit Copy Delete	15	21922	152	2005-07-22	F
Edit Copy Delete	16	8633	4982	1993-01-03	F
Edit Copy Delete	17	11980	9184	2008-06-14	F
Edit Copy Delete	18	1089	5883	1972-03-30	D
Edit Copy Delete	19	5155	1422	1985-04-05	A
Edit Copy Delete	20	10566	4460	1999-01-30	B
Edit Copy Delete	21	17854	2800	2005-04-22	D
Edit Copy Delete	22	31031	7200	1989-03-15	C
Edit Copy Delete	23	13047	3156	1985-05-08	A
Edit Copy Delete	24	1779	3134	1992-07-07	A
Edit Copy Delete	25	2122	6104	2022-02-10	A
Edit Copy Delete	26	4371	1252	2012-04-29	C
Edit Copy Delete	27	4923	7915	1998-11-18	D
Edit Copy Delete	28	11284	9815	1993-12-27	D
Edit Copy Delete	29	23202	1505	1988-06-06	C
Edit Copy Delete	30	27726	7813	1985-02-08	F
Edit Copy Delete	31	37437	8676	2016-02-09	B
Edit Copy Console Delete	32	2006	5744	2001-04-14	C
Edit Copy Delete	33	2664	5212	2018-07-16	D
Edit Copy Delete	34	33226	5014	2001-12-24	C
Edit Copy Delete	35	39230	6814	2007-10-18	D
Edit Copy Delete	36	27994	7929	1983-12-16	C
Edit Copy Delete	37	28965	2575	1988-05-26	A
Edit Copy Delete	38	32735	7837	2003-08-08	A
Edit Copy Delete	39	14898	394	2008-10-19	A
Edit Copy Delete	40	16045	6073	1982-10-12	A
Edit Copy Delete	41	39302	1399	1978-11-13	B
Edit Copy Delete	42	39242	4218	1970-11-28	D
Edit Copy Delete	43	35246	8328	1970-06-23	F
Edit Copy Delete	44	5189	6958	2004-09-04	A
Edit Copy Delete	45	36434	7040	1981-08-21	B
Edit Copy Delete	46	22446	6424	2021-12-23	D
Edit Copy Delete	47	15268	355	1997-09-21	D
Edit Copy Delete	48	35448	7706	2009-09-04	D
Edit Copy Delete	49	18898	494	1990-10-11	A
Edit Copy Delete	50	10856	4253	1998-03-23	A
Edit Copy Console Delete	51	6760	4453	1994-10-05	A
Edit Copy Delete	52	12256	7776	1993-04-19	D
Edit Copy Delete	53	2767	7561	2024-08-23	A
Edit Copy Delete	54	11882	8918	2005-04-10	A
Edit Copy Delete	55	27909	4609	1983-05-03	D
Edit Copy Delete	56	18999	134	1973-02-24	A
Edit Copy Delete	57	33169	4331	1983-07-23	C
Edit Copy Delete	58	24902	6416	1972-08-06	D
Edit Copy Delete	59	17887	1855	2014-08-19	B
Edit Copy Delete	60	25954	7065	2009-07-07	A
Edit Copy Delete	61	33319	3073	2003-07-31	C
Edit Copy Delete	62	7688	8335	2019-01-23	D
Edit Copy Delete	63	4661	9444	2002-12-28	C
Edit Copy Delete	64	16097	6231	2017-06-12	D
Edit Copy Delete	65	21670	6305	1999-04-01	D
Edit Copy Delete	66	35913	7681	1994-01-23	B
Edit Copy Delete	67	29618	895	2016-05-18	C
Edit Copy Delete	68	1126	3389	1981-06-25	C
Edit Copy Delete	69	38697	1612	2006-03-07	D
Edit Copy Console Delete	70	35497	6909	2022-11-02	A

←T→	enrollment_id	studentID	courseID	enrollmentDate	grade
Edit Copy Delete	71	21713	3098	1973-06-08	A
Edit Copy Delete	72	26983	5038	2013-08-01	A
Edit Copy Delete	73	13301	3221	1978-03-14	F
Edit Copy Delete	74	23048	2202	1973-01-26	C
Edit Copy Delete	75	19918	6535	2001-11-08	B
Edit Copy Delete	76	23287	8702	1979-05-11	B
Edit Copy Delete	77	18843	6742	1971-08-12	A
Edit Copy Delete	78	32792	3255	2013-01-26	B
Edit Copy Delete	79	36221	2881	1993-06-09	D
Edit Copy Delete	80	15268	8329	2023-08-11	B
Edit Copy Delete	81	25113	7859	1997-08-22	B
Edit Copy Delete	82	2423	8335	1998-12-06	A
Edit Copy Delete	83	8006	6988	1984-08-29	F
Edit Copy Delete	84	8719	8908	1977-04-20	B
Edit Copy Delete	85	6503	2835	1974-02-04	C
Edit Copy Delete	86	16470	3246	1995-01-11	B
Edit Copy Delete	87	9248	9542	1983-11-25	D
Edit Copy Delete	88	30519	7993	2002-04-26	F
Edit Copy Delete	89	37634	5734	1991-09-02	A
Edit Copy Delete	90	18138	2894	2016-05-12	C
Edit Copy Delete	91	21482	4835	2003-11-30	F
Edit Copy Delete	92	30097	8423	1998-06-18	D
Edit Copy Delete	93	14169	193	2003-08-01	B
Edit Copy Delete	94	3528	7010	2021-11-21	D
Edit Copy Delete	95	6974	4901	1983-12-27	F
Edit Copy Delete	96	5487	5047	2005-12-31	F
Edit Copy Delete	97	8209	5215	1975-08-01	F
Edit Copy Delete	98	35319	4485	1981-01-02	B
Edit Copy Delete	99	33153	6395	2017-08-17	A
Edit Copy Delete	100	20877	3739	2004-10-05	D

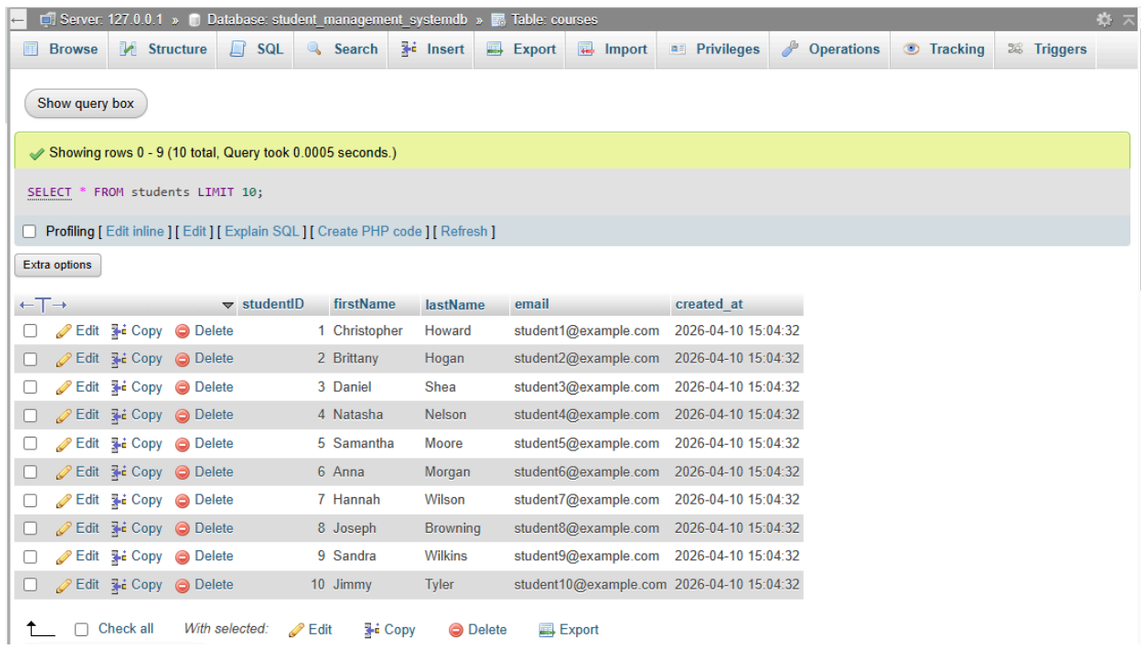
Result:
Queries executed successfully with acceptable response time, demonstrating that the database performs efficiently even with large volumes of data.

6. **Sample Data Validation**
Objective: Verify that inserted data appears realistic and correctly formatted.

SQL Commands Used:

```
SELECT * FROM students LIMIT 10;
SELECT * FROM courses LIMIT 10;
```

Screenshots:



studentID	firstName	lastName	email	created_at
1	Christopher	Howard	student1@example.com	2026-04-10 15:04:32
2	Brittany	Hogan	student2@example.com	2026-04-10 15:04:32
3	Daniel	Shea	student3@example.com	2026-04-10 15:04:32
4	Natasha	Nelson	student4@example.com	2026-04-10 15:04:32
5	Samantha	Moore	student5@example.com	2026-04-10 15:04:32
6	Anna	Morgan	student6@example.com	2026-04-10 15:04:32
7	Hannah	Wilson	student7@example.com	2026-04-10 15:04:32
8	Joseph	Browning	student8@example.com	2026-04-10 15:04:32
9	Sandra	Wilkins	student9@example.com	2026-04-10 15:04:32
10	Jimmy	Tyler	student10@example.com	2026-04-10 15:04:32

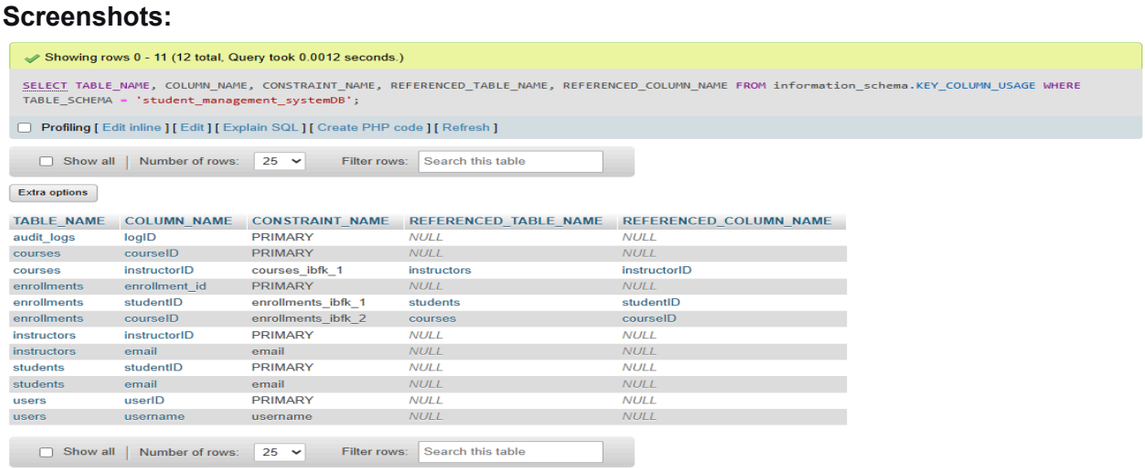
Result:
Sample records show properly formatted and realistic data. This confirms that the dataset is valid.

7. **Relationship (ERD lines) verification**
Objective: Verify that all relationships defined in the ERD are properly implemented in the database schema using foreign keys.

SQL Commands Used:

```
SELECT
    TABLE_NAME,
    COLUMN_NAME,
    CONSTRAINT_NAME,
    REFERENCED_TABLE_NAME,
    REFERENCED_COLUMN_NAME
FROM information_schema.KEY_COLUMN_USAGE
WHERE TABLE_SCHEMA = 'student_management_systemDB';
```

Screenshots:



TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_TABLE_NAME	REFERENCED_COLUMN_NAME
audit_logs	logID	PRIMARY	NULL	NULL
courses	courseID	PRIMARY	NULL	NULL
courses	instructorID	courses_ibfk_1	instructors	instructorID
enrollments	enrollment_id	PRIMARY	NULL	NULL
enrollments	studentID	enrollments_ibfk_1	students	studentID
enrollments	courseID	enrollments_ibfk_2	courses	courseID
instructors	instructorID	PRIMARY	NULL	NULL
instructors	email	email	NULL	NULL
students	studentID	PRIMARY	NULL	NULL
students	email	email	NULL	NULL
users	userID	PRIMARY	NULL	NULL
users	username	username	NULL	NULL

Result:
The query successfully displayed all foreign key constraints in the database. It confirms that all relationships defined in the ERD are correctly implemented in the schema. Each relationship between tables (such as students to enrollments and courses to enrollments) is properly enforced using foreign keys, ensuring referential integrity and consistency across the database.